Windows Firewall Management

Implementation Solution Guide

Version 1.0.0 · 2026-06-13

Contents

Executive Summary — Windows Firewall Management (Microsoft Intune) 4

What you get . . . 4

What this delivers . . . 4

The standard deployment model . . . 4

Outcome . . . 4

1. Solution Overview — Windows Firewall Management 4

What this is . . . 4

How it works . . . 5

What you can manage . . . 5

Benefits . . . 5

Limitations (be honest with buyers) . . . 5

The standard deployment model . . . 5

2. Requirements Guide — Windows Firewall Management 5

2.1 Licensing . . . 6

2.2 Device enrollment . . . 6

2.3 Supported Windows OS . . . 6

2.4 Inventory of legitimate inbound needs . . . 6

2.5 No conflicting firewall management . . . 6

Checklist . . . 6

Global Requirements — Intune Solutions (category-level) 6

G1. Microsoft Entra ID tenant . . . 7

G2. Microsoft Intune subscription . . . 7

G3. Administrative roles . . . 7

G4. Device enrollment in Intune . . . 7

G5. Pilot security group . . . 7

G6. Supported Windows baseline . . . 7

G7. Network access to Microsoft cloud endpoints . . . 7

3. Implementation Guide — Windows Firewall Management 8

Step 1 — Create the pilot group . . . 8

Step 2 — Create the firewall policy . . . 8

Step 3 — Configure the firewall settings . . . 8

Step 4 — Assign to the pilot group . . . 8

Step 5 — (Optional) create a firewall rules policy . . . 8

Step 6 — Apply on a pilot device . . . 9

Step 7 — Validate the pilot . . . 9

Step 8 — Expand to production . . . 9
5. Testing Guide — Windows Firewall Management 9
Test 1 — Policy applied . . . 9
Test 2 — All profiles enabled . . . 9
Test 3 — Default inbound is Block . . . 9
Test 4 — Legitimate inbound still works . . . 10
Test 5 — Rules applied (if a rules policy is used) . . . 10
Test 6 — No conflicts . . . 10
Test 7 — Compliance intact . . . 10
Pass/fail gate . . . 10
6. Troubleshooting Guide — Windows Firewall Management 10
A. Firewall not enabled / profiles off . . . 10
B. Legitimate inbound blocked . . . 10
C. Conflicts . . . 11
D. Rules not taking effect . . . 11
E. Diagnostics quick reference . . . 11
F. When to escalate . . . 11
7. Best Practices — Windows Firewall Management 11
8. Deliverables Summary — Windows Firewall Management 12
Documentation . . . 12
PowerShell scripts . . . 12
What the buyer receives . . . 13
Outcome . . . 13
Appendix A — PowerShell Scripts 14

Executive Summary — Windows Firewall Management (Microsoft Intune)

White-label · Intune Solutions

The host firewall is the last line of defense on every endpoint — but on unmanaged fleets it's often disabled, inconsistently configured, or full of stale rules. **Windows Firewall Management** through Microsoft Intune enforces the firewall on across every network profile, sets a sane **default-deny inbound** posture, deploys standardized rules, and reports the firewall state of every device.

This is a complete, **deploy-it-today** package for managing Windows Defender Firewall with Microsoft Intune, written for an administrator with no prior firewall-management experience.

What you get

- A plain-language **Solution Overview** of Intune-managed firewall policy.
- A **Requirements Guide** with verify / implement-if-missing for every prerequisite.
- An **Implementation Guide** of precise, click-by-click Intune steps in the required order.
- Five **PowerShell scripts** for firewall discovery, validation, rule reporting, remediation, and conflict checking.
- **Testing** and **Troubleshooting** guides for confident rollout.

What this delivers

An Intune **firewall policy** that enables the firewall for **Domain, Private, and Public** profiles, sets **default inbound = Block / outbound = Allow**, configures logging and stealth mode, and (optionally) deploys a standardized **firewall rules** set — assigned to a pilot, then expanded.

The standard deployment model

Assess → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

A consistently enforced host firewall across the fleet with default-deny inbound and centralized rules — deployable by a single administrator and validated before rollout.

1. Solution Overview — Windows Firewall Management

What this is

Windows Defender Firewall is the built-in host firewall on every Windows device. This solution manages it centrally through Microsoft Intune: enforcing it **on** for all network profiles, setting default inbound/outbound behavior, deploying standardized firewall **rules**, and reporting each device's firewall state — so the host firewall is consistent and auditable across the fleet.

How it works

1. An Intune **firewall policy** (Endpoint security → Firewall) sets the firewall **on** for the **Domain, Private, and Public** profiles.
1. It defines **default actions** — typically **inbound = Block**, **outbound = Allow** — plus logging, stealth mode, and unicast-response behavior.
1. An optional **firewall rules** policy deploys specific allow/block rules (ports, apps, IP scopes).
2. Intune applies and **reports** the firewall state; devices that drift are visible.

What you can manage

- **Firewall enabled** per profile (Domain/Private/Public).
- **Default inbound/outbound** actions.
- **Logging** (dropped packets, successful connections) and log size/path.
- **Stealth mode**, **shielded** mode, unicast response to multicast/broadcast.
- **Custom firewall rules** (allow/block by port, program, protocol, or address).

Benefits

- **Consistent enforcement** — the firewall is on everywhere, no exceptions.
- **Default-deny inbound** — reduces exposure to lateral movement and scanning.
- **Centralized rules** — one managed rule set instead of per-device drift.
- **Visibility** — fleet-wide firewall-state reporting for compliance and audits.

Limitations (be honest with buyers)

- Firewall settings can **conflict** with a legacy GPO firewall or the security baseline — manage in one place.
- Overly strict rules can block legitimate apps — pilot and validate first.
- Custom rules must be maintained as your app/port landscape changes.
- Some specialized scenarios (server roles, IPsec connection security rules) need extra design beyond this baseline.

The standard deployment model

> ***Create the firewall policy (on + default-deny inbound)*** → ***(optional) deploy rules*** → ***assign to a pilot*** → ***validate*** → ***expand.*** *repeatable pattern:* ***Assess*** → ***Configure*** → ***Pilot*** → ***Validate*** → ***Expand.***

2. Requirements Guide — Windows Firewall Management

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in `global-requirements-intune-solutions.md`.

2.1 Licensing

Description: Intune (Plan 1). **Why:** Intune delivers and reports the firewall policy. **Verify:** M365 admin center → **Billing** → **Licenses**. **Implement:** Assign M365 E3/E5 or Business Premium (Intune included).

2.2 Device enrollment

Description: Devices enrolled in Intune. **Why:** The firewall policy applies only to enrolled devices. **Verify:** Intune → **Devices** → **All devices**. **Implement:** Enable automatic enrollment; enroll devices.

2.3 Supported Windows OS

Description: Windows 10/11. **Why:** The firewall CSP settings require a supported Windows build. **Verify:** `winver`; run `WFM-Discovery.ps1`. **Implement:** Patch/upgrade to a supported build.

2.4 Inventory of legitimate inbound needs

Description: Awareness of apps/services that require inbound connections. **Why:** Default-deny inbound can block legitimate listeners (e.g. Remote Help, remote management, line-of-business servers). **Verify:** Review current inbound rules (`WFM-Discovery.ps1` lists rule counts; export details as needed). **Implement:** Plan explicit allow rules for required inbound services.

2.5 No conflicting firewall management

Description: No legacy GPO firewall or baseline managing the same settings. **Why:** Two policies setting the firewall fight each other. **Verify:** Run `WFM-ConflictCheck.ps1`; check the security baseline for firewall settings. **Implement:** Make Intune the single firewall authority; disable/scope away the GPO and baseline firewall settings.

Checklist

- Intune licensing assigned
- Devices enrolled
- Supported Windows build — `WFM-Discovery.ps1`
- Legitimate inbound needs documented (allow rules planned)
- No conflicting GPO/baseline firewall management — `WFM-ConflictCheck.ps1`
- Pilot Entra group created

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation Guide — Windows Firewall Management

Follow these steps **in order**, in the **Microsoft Intune admin center** (`https://intune.microsoft.com`).

> ***Model:*** *A firewall policy enabling all profiles with default-deny inbound, plus an optional rules policy, assigned to a pilot group.*

Step 1 — Create the pilot group

1. Entra → **Groups** → **New group** → Security → `Firewall-Pilot` → add pilot devices. **Create**.

Step 2 — Create the firewall policy

1. Intune → **Endpoint security** → **Firewall** → **Create Policy**.
2. **Platform:** Windows. **Profile: Windows Firewall**. **Create**.
3. **Name:** `Firewall - Pilot`. **Next**.

Step 3 — Configure the firewall settings

On **Configuration settings**, for **Domain**, **Private**, and **Public** profiles set:

1. **Firewall enabled** = **Allowed** (turns the firewall on for that profile).
2. **Default inbound action** = **Block**.
3. **Default outbound action** = **Allow**.
4. **Stealth mode** = **Block** (not configured to disable) — keeps the device from responding to probes.
5. **Inbound notifications** = **Block** (suppress per-app prompts to users) or as your policy prefers.
6. **Unicast responses to multicast broadcast** = **Block** (recommended).
7. **Log file path / size**, **Log dropped packets** = **Enable**, **Log successful connections** = optional.

Click **Next**.

Step 4 — Assign to the pilot group

1. **Assignments** → **Add groups** → `Firewall-Pilot`. **Next** → **Create**.

Step 5 — (Optional) create a firewall rules policy

Only if you need explicit allow/block rules (e.g. allow Remote Help inbound, allow a LOB app).

1. **Endpoint security** → **Firewall** → **Create Policy** → **Windows** → **Windows Firewall Rules**. **Create**.
2. **Name:** `Firewall Rules - Pilot`. Add each rule: **Name**, **Direction** (In/Out), **Action** (Allow/Block), **Protocol/Ports**, **Program**, **Local/Remote address** scope.

3. **Assignments** → **Add groups** → `Firewall-Pilot`. **Create**.

Step 6 — Apply on a pilot device

1. On a pilot device, sync policy (**Settings** → **Accounts** → **Access work or school** → **Info** → **Sync**).
2. Confirm the firewall is on for all profiles and the default inbound action is Block.

Step 7 — Validate the pilot

1. Run `WFM-Validation.ps1` → confirms all profiles enabled with default-deny inbound.
2. Run `WFM-RuleReport.ps1` → lists effective firewall rules.
3. Run `WFM-ConflictCheck.ps1` → flags GPO/baseline firewall conflicts. Complete the **Testing Guide**.

Step 8 — Expand to production

1. Add a production group to the firewall policy **Assignments**; roll out in waves.
2. Verify required inbound services still work (add explicit allow rules as needed) before broad rollout.
3. Re-validate each wave with `WFM-Validation.ps1`.

Order of operations

```
Pilot group → Firewall policy (profiles on + default-deny inbound) →
(optional) rules policy → Assign → Apply → Validate → Expand
```

5. Testing Guide — Windows Firewall Management

Run after the pilot policy applies (Implementation Step 7) and after each production wave.

Test 1 — Policy applied

Do: Intune → **Endpoint security** → **Firewall** → `Firewall - Pilot` → **Device status**. **Expected:** Pilot devices show **Succeeded**.

Test 2 — All profiles enabled

Do: On a pilot device run `WFM-Validation.ps1` or `Get-NetFirewallProfile`. **Expected:** Domain, Private, and Public profiles all **Enabled = True**.

Test 3 — Default inbound is Block

Do: `Get-NetFirewallProfile | Select Name, DefaultInboundAction, DefaultOutboundAction`. **Expected: DefaultInboundAction = Block**, **DefaultOutboundAction = Allow** for all profiles.

Test 4 — Legitimate inbound still works

Do: Exercise required inbound services (e.g. Remote Help, remote management) on a pilot device. **Expected:** Allowed services still work (via explicit allow rules); everything else inbound is blocked.

Test 5 — Rules applied (if a rules policy is used)

Do: Run `WFM-RuleReport.ps1`. **Expected:** The deployed allow/block rules are present and effective.

Test 6 — No conflicts

Do: Run `WFM-ConflictCheck.ps1`; review Intune device → firewall policy status. **Expected:** No GPO/baseline policy contesting the firewall settings.

Test 7 — Compliance intact

Do: Intune → **Devices** → **Compliance** (with a firewall compliance rule, if used). **Expected:** Devices remain **Compliant**; the firewall requirement passes.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1–4 pass; required inbound services confirmed working
Wave pass	Tests 2–3, 7 pass before the next wave

6. Troubleshooting Guide — Windows Firewall Management

Use `WFM-Validation.ps1` / `WFM-Discovery.ps1` for firewall state and `WFM-RuleReport.ps1` / `WFM-ConflictCheck.ps1` to inspect rules and conflicts, `WFM-Remediation.ps1` for safe fixes.

A. Firewall not enabled / profiles off

Symptom	Cause	Remediation
A profile shows Disabled	Policy not applied / synced, or a conflicting policy disabled it	Run `WFM-Remediation.ps1` (re-enables + syncs); resolve conflicts (`WFM-ConflictCheck.ps1`)
Policy "Error"	Device not synced/enrolled	Sync; confirm device in `Firewall-Pilot`

B. Legitimate inbound blocked

Symptom	Cause	Remediation
Remote Help / remote mgmt / LOB app can't connect inbound	Default-deny inbound with no allow rule	Add an explicit **allow inbound** rule (rules policy) for the required port/program; re-test
App worked before, now blocked	Default inbound flipped to Block	Expected — create the allow rule for it; this is the point of default-deny

C. Conflicts

- A legacy **GPO firewall** or the **security baseline** also sets firewall state → they fight Intune. Make Intune the single authority; disable/scope the GPO and remove firewall settings from the baseline. Use `WFM-ConflictCheck.ps1`.
- "Merge" behavior: local rules may still apply unless you disable local rule merge — decide whether local admins can add rules.

D. Rules not taking effect

- Rule present but not working → check direction/protocol/port/scope; confirm the profile the rule applies to matches the active network profile.
- Use `WFM-RuleReport.ps1` to see effective rules and their enabled state.

E. Diagnostics quick reference

```
Get-NetFirewallProfile | Select Name, Enabled, DefaultInboundAction, DefaultOutboundAction
Get-NetFirewallRule -Enabled True | Select DisplayName, Direction, Action | Sort Direction
netsh advfirewall show allprofiles
```

Firewall log (if enabled): `%systemroot%\system32\LogFiles\Firewall\pfirewall.log`.

F. When to escalate

If profiles are enabled, the policy shows Succeeded, no conflicts are reported, and a required service is still blocked with an allow rule in place, capture `Get-NetFirewallRule` output and the firewall log and open a Microsoft support case.

7. Best Practices — Windows Firewall Management

- **Enable all three profiles.** Domain, Private, and Public should all have the firewall on — no exceptions.
- **Default-deny inbound, allow outbound.** The single highest-value posture; add explicit allow rules only for required inbound services.
- **Pilot before fleet.** Default-deny inbound can block legitimate listeners — confirm required services work on a pilot first.

• **Inventory inbound needs first.** Know what must accept inbound (Remote Help, remote management, LOB servers) and pre-build those allow rules.

• **One firewall authority.** Manage the firewall via Intune only — retire the GPO firewall and remove firewall settings from the security baseline to avoid conflicts.

• **Decide on local rule merge.** Choose whether local admins can add rules; for locked-down fleets, disable local merge.

• **Enable logging.** Log dropped packets to diagnose blocked legitimate traffic quickly.

• **Use stealth mode.** Keep devices from responding to unsolicited probes.

• **Name rules clearly.** `Allow-Inbound-<service>-<port>` so intent is obvious in audits.

• **Report regularly.** Track firewall state across the fleet with `WFM-RuleReport.ps1` / Intune reporting.

8. Deliverables Summary — Windows Firewall Management

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	What Intune-managed firewall is, how it works, benefits, limitations
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Intune Solutions)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered Intune steps (profiles, defaults, rules)
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Field issues and rule handling
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`WFM-Discovery.ps1`	Reports firewall profile state and rule counts on the device
`WFM-Validation.ps1`	Confirms all profiles enabled with default-deny inbound
`WFM-RuleReport.ps1`	Lists effective firewall rules (direction/action/port)
`WFM-Remediation.ps1`	Safe remediation; re-enables firewall profiles and forces a sync
`WFM-ConflictCheck.ps1`	Lists Intune firewall policies/assignments to spot conflicts (Microsoft Graph)

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process following the repeatable model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

A consistently enforced host firewall across the fleet with default-deny inbound and centralized rules.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

WFM-ConflictCheck.ps1

WFM-Discovery.ps1

WFM-Remediation.ps1

WFM-RuleReport.ps1

WFM-Validation.ps1